Addressing Stochastic Rising Bandits with Thompson Sampling

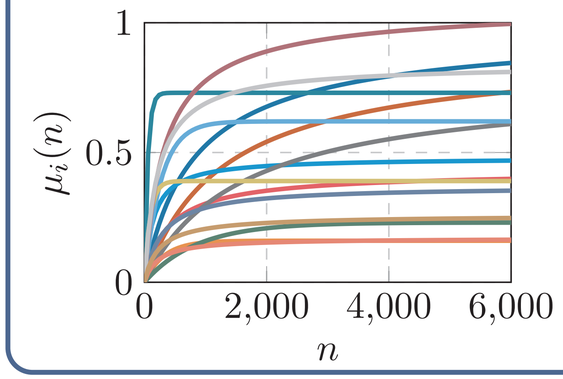
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Setting - Overview

Stochastic Rising Rested Bandits

- Samples of arm i taken from either a Bernoulli or a σ^2-subgaussian distribution with expectation $\mu_i(n)$, being n the number of pulls of the arm.
- Goal: minimize the *expected cumulative regret*:

$$R(\mathfrak{A}, T) := T\,\overline{\mu}_{i^*(T)}(T) - \mathbb{E}\left[\sum_{t=1}^{T} \mu_{I_t}(N_{I_t,t})\right]$$



Motivation

- Real-world applications present scenarios where the rewards associated with each arm evolve.
- We study the domain of *stochastic rising rested bandits* (SRRB, Heidari et al., 2016; Metelli et al., 2022).
- The SRRB models situations where the arms' expected rewards *increase* with the pulls, describing one of the most fundamental settings in Automatic Machine Learning (AutoML).
- Bayesian methods, which were notoriously hard to study, lacked an analysis in this setting, providing excellent experimental results (Metelli et al. (2022)).

Contributions

- Analysis of Thompson Sampling variants with forced exploration: `ET-Beta-TS` and its Gaussian extension γ-`ET-GTS`, with novel analytical techniques.
- A lower bound dependent on a new complexity measure.
- Sliding-window extensions (`ET-Beta-SWTS`, γ-`ET-SWGTS`) for limited horizon T.

Bounding the Online Regret

Regret for ET-Beta-TS:

$$\mathbb{E}[N_{i,T}] \leqslant O\left(\Gamma + \boxed{(1+\epsilon)\frac{\log(T)}{d(\overline{\mu}_i(T), \overline{\mu}_1(\sigma))} + \frac{1}{\epsilon^2}} + \boxed{\sum_{j=\Gamma}^{\sigma} \frac{1}{(1-\overline{\mu}_1(\sigma))^{j+1}}\delta_{\text{TV}}\left(\text{Bin}(j, \underline{\mu}_1(j)), \text{Bin}(j, \overline{\mu}_1(\sigma))\right)}\right)$$

For $\Gamma \geqslant \sigma$:

$$\mathbb{E}[N_{i,T}] \leqslant O\left(\Gamma + \frac{\log(T)}{\Delta_i^2}\right)$$

The Sample Complexity of the Rising Rested Problem:

$\mathcal{M}_{\overline{\sigma}} := \{\boldsymbol{\mu}\ \text{SRRB} \ : \sigma \leqslant \overline{\sigma}\}$. Let $T \in \mathbb{N}$ and $\overline{\sigma} \in [\![2, \frac{T-1}{2}]\!]$. For every algorithm $\mathfrak{A}$, it holds that:

$$\sup_{\boldsymbol{\mu} \in \mathcal{M}_{\overline{\sigma}}^{\text{det}}} R_{\boldsymbol{\mu}}(\mathfrak{A}, T) \geqslant \frac{K}{64}(\overline{\sigma} - 2).$$

Thus, *any algorithm* that wishes to be no-regret, cannot avoid playing every arm a number of times proportional to $\overline{\sigma}$, suggesting that for $\Gamma \propto \overline{\sigma}$ we are tight.

Facing the limited Time Horizon problem-Forgetting the past:

$$\mathbb{E}[N_{i,T}] \leqslant O\left(\Gamma + \boxed{\frac{T}{\tau}}\boxed{\frac{\log(T)}{(\Delta_i'(T;\tau))^2}} + \boxed{(\sigma'(T;\tau) - \Gamma)\tau}\right)$$

For $\tau\sqrt{\sigma'(T;\tau)} = \Theta(\sqrt{T\log(T)})$, even when $\Gamma = 0$:

$$R(\texttt{Beta-SWTS}) \leqslant O\left(\frac{1}{\Delta^2}\sqrt{\sigma'(T;\tau)T\log(T)}\right)$$

Novelty: Concentration Properties for Beta-TS in Non-Stationary Settings

Beta distribution does not enjoy nice concentration properties like the Gaussian distribution.

Bounding a ubiquitous term in TS analysis for Non-Stationary Settings:

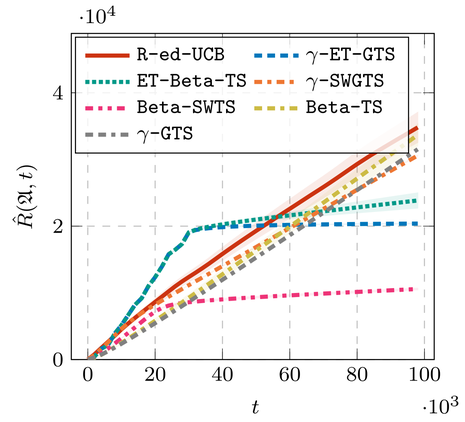
Being $p_{i,t} := \Pr\left(\text{Beta}\left(S_{1,t}+1, F_{1,t}+1\right) > y_i | \mathcal{F}_{t-1}\right).$

$$\mathbb{E}_{S_{1,t}\sim\text{PB}(\underline{\mu}_1(j))}\left[\frac{1}{p_{i,t}} \,\middle|\, N_{1,t} = j\right] \leqslant \mathbb{E}_{S_{1,t}\sim\text{Bin}(j,\overline{\mu}_1(j))}\left[\frac{1}{p_{i,t}} \,\middle|\, N_{1,t} = j\right] \leqslant \mathbb{E}_{S_{1,t}\sim\text{Bin}(j,x)}\left[\frac{1}{p_{i,t}} \,\middle|\, N_{1,t} = j\right],$$

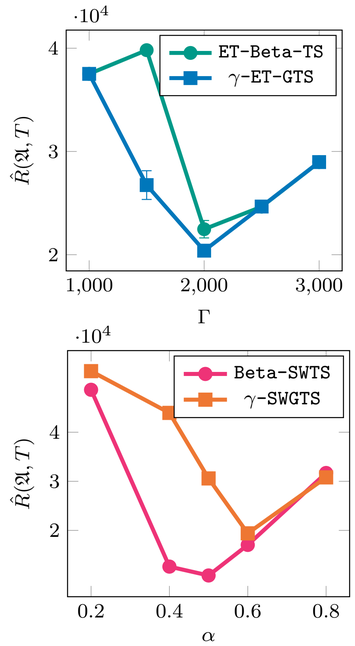
These results state that under the right circumstances those concentration properties are the same of those already retrieved in the non-stationary literature that can now be leveraged to expand the analysis of variants of Thompson Sampling to a large variety of evolving environments, making possible to widen results for rested and restless scenarios.

Experimental Validation

Online Regret



Sensitivity Analysis



References

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